

1 Preliminaries

This section establishes the foundational medical, computational, and mathematical concepts required for automated leukemia detection and classification.

1.1 Medical Foundations and Leukemia Taxonomy

Leukemia is a hematological malignancy characterized by the abnormal proliferation of White Blood Cells (WBCs). These malignant cells originate in the bone marrow and interfere with the production of healthy blood cells.

Let

$$\mathcal{C} = \{c_1, c_2, \dots, c_K\}$$

denote the set of classes in this study, where:

- c_1 (**Healthy**): Normal WBC morphology.
- c_2 (**ALL** – Acute Lymphoblastic Leukemia): Rapidly progressing; affects lymphoid cells.
- c_3 (**AML** – Acute Myeloid Leukemia): Aggressive; affects myeloid lineage cells.
- c_4 (**CLL** – Chronic Lymphocytic Leukemia): Slow-growing; affects mature lymphocytes.
- c_5 (**CML** – Chronic Myeloid Leukemia): Characterized by uncontrolled growth of myeloid cells.

Peripheral Blood Smear (PBS): PBS is the gold-standard diagnostic tool in which blood is spread on a glass slide and stained. It enables visual analysis of WBC morphology, including cell size, nuclear shape, and cytoplasmic texture.

1.2 Image Preprocessing and Data Preparation

To enhance data quality and ensure model compatibility, a transformation $T(\cdot)$ is applied to raw images

$$X \in \mathbb{R}^{H \times W \times 3}$$

such that:

$$X' = T(X)$$

Preprocessing: Includes resizing for fixed model input dimensions, Gaussian blur for noise reduction, and Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance nuclear details.

Data Augmentation: To mitigate overfitting in limited medical datasets, N synthetic samples are generated:

$$\tilde{X}_i = A_i(X), \quad i = 1, 2, \dots, N$$

where $A_i(\cdot)$ represents stochastic transformations such as rotation, flipping, and zooming.

Class Imbalance: Occurs when

$$N_{\text{healthy}} \gg N_{\text{leukemic}}$$

This imbalance is addressed using Synthetic Minority Over-sampling Technique (SMOTE), random oversampling, or class weighting w_k in the loss function.

1.3 Machine Learning Framework

The framework employs Convolutional Neural Networks (CNNs) for automated feature extraction. The output of the l -th convolutional layer is given by:

$$\mathbf{F}^{(l)} = \sigma(\mathbf{W}^{(l)} * \mathbf{F}^{(l-1)} + \mathbf{b}^{(l)})$$

where $*$ denotes convolution, $\sigma(\cdot)$ is an activation function, and $\mathbf{F}^{(l)}$ represents the feature map.

Hierarchical Learning: CNNs progressively learn low-level edge features and high-level morphological patterns relevant to leukemia detection.

Transfer Learning: Model parameters θ are initialized using pre-trained weights (e.g., ResNet or MobileNet):

$$\theta = \theta_{\text{pretrained}} + \Delta\theta$$

This accelerates convergence and improves generalization.

Explainable AI (XAI): Gradient-weighted Class Activation Mapping (Grad-CAM) is used to generate heatmaps highlighting regions of interest (ROI) that influence model predictions, enhancing clinical interpretability.

1.4 Loss Function and Performance Metrics

To address multi-class classification and class imbalance, the Weighted Categorical Cross-Entropy loss is used:

$$\mathcal{L} = - \sum_{k=1}^K w_k y_k \log(\hat{y}_k)$$

where y_k and $\hat{y}_k$ denote the ground truth and predicted probability for class k , respectively.

Model performance is evaluated using a confusion matrix comprising True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

Table 1: Performance Evaluation Metrics

Metric	Formula	Description
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall correctness
Precision	$\frac{TP}{TP+FP}$	Reliability of leukemia detection
Recall	$\frac{TP}{TP+FN}$	Sensitivity to leukemia cases
F1-score	$2 \cdot \frac{PR \cdot RE}{PR+RE}$	Balance between precision and recall
G-Mean	$\sqrt{\text{Sensitivity} \times \text{Specificity}}$	Performance on imbalanced data